Community opposition to data centers in cities is clear and growing

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Abstract

Rapid growth in computing and generative artificial intelligence is driving unprecedented demand for data center construction. We provide the first large-scale analysis of how U.S. cities discuss and respond to data centers using transcripts of local government meetings from 2009–2026. We find that debates center on tradeoffs between economic development and perceived local costs, and that residents are consistently less supportive of data centers than city officials.

1 Main

Data centers currently account for around 1.5% of global electricity consumption, and that rate is expected to more than double by 2030 [1]. Political and economic pressures to meet this demand are high, as the transformative potential of artificial intelligence depends largely on access to computational power offered by data centers [2].

Despite their global impacts, data centers are infrastructure projects that must be sited in local cities. Decisions about data center construction matter for a broad set of local policy areas—data centers compete with residential, commercial, and industrial development for land use; they impose concentrated demands on water, electricity, and road infrastructure; and decisions about property taxes can have large implications for local tax revenues. City officials are responsible for many decisions related to building and siting data centers and must weigh these complicated new projects alongside existing priorities. Recent media accounts have documented how debates over new data center construction can become contentious and polarizing [3–5].

Yet, beyond the key insights offered by these accounts about specific places and projects, we have little broad evidence about how cities are systematically grappling with data centers. Existing work on land use in political science—most often on housing development—suggests residents who have strong feelings about issues like data centers are likely to turn to public meetings to express their views directly to city officials [6–8]. Further, work on political

representation suggests that—in general—policies adopted by cities tend to align with the preferences of residents [9]. But, this relationship is hardly straightforward: not all residents are equally likely to have their preferences reflected in local policy [10], due in part to how institutional design influences the incentives that city officials have to formalize constituent preferences into policy [e.g. 11–14].

We offer the first large-scale evidence on how people in cities across the United States are discussing data centers in their local governments. Although data center construction is growing rapidly, the decentralized nature of American government means that there is no easily accessible data on how people in cities are discussing data centers [15, 16]. We address this lack of data by extending *LocalView*—the largest publicly available database of local government meetings—to systematically collect nearly 150,000 transcripts of city meetings in 48 states from 2009-2026 [17]. Our goal is not to approximate a representative survey to measure public opinion on data centers, several of which are already available (For example, see here). Instead, we study how data center discussions unfold in actual local governance deliberation among both elected officials and residents. We use these transcripts to (1) characterize the extent of discussion about data centers in cities over time, (2) identify the most commonly discussed issues, (3) identify the tone in which people discuss those issues, and (4) distinguish speakers as members of the public or city officials.

First, we find that local government attention to data centers has dramatically increased in recent years. The left plot of Figure 1 shows the proportion of all local meetings in three month bins that discuss data centers. Since the beginning of 2023, we find local government attention to data centers (shown in maize) has increased by more than a factor of 10. We benchmark this rise in attention to data centers against other examples of projects commonly referred to as "locally unwanted land use"—casinos, drug treatment clinics, and sports arenas. While attention to other issues has remained relatively flat since 2023, attention to data centers has skyrocketed with no signs of slowing.

We identify 1,707 meetings that discuss data centers and use text analysis to assign whether or not each discussion of data centers concerns community costs (e.g. safety, noise, traffic), economic development, land use and infrastructure, utilities and the environment, and/or data center projects in other cities. In Supplementary Figure 1, we show that these conversations are concentrated—yet do not occur exclusively—in cities with existing or proposed data centers.

Next, the right plot of Figure 1 shows that land use and infrastructure, electricity, water, and other utilities are among the topics most commonly mentioned in discussions of data centers (appearing in 25–40% of data center discussions). Other tangible concerns about noise and traffic are less common but still appear often (10–15%). Nearly all commentary related to economic development—most often related to jobs and tax benefits—appear around the middle of our distribution (20–30%). These patterns reinforce that, at their core, discussions about data center development grapple with how cities allocate space and resources, and who will bear the brunt of associated costs.

Next, we use tone classifications described in *Methods* to show whether speakers who discuss each topic tend to support or oppose data centers in their communities. For each topic, we calculate the difference between the proportion of speakers who invoked it in positive and negative ways (we drop neutral passing references). The left plot of Figure 2 shows how speakers tend to be more positive when discussing the potential economic

impacts of data centers, such as the prospect of new jobs. Economic development is the most positive theme associated with data centers. Most other concerns are much more negative. We find this support pattern is largely asymmetric—speakers invoke more topics in a primarily negative way than a primarily positive way.

Finally, we show that opposition to data centers is driven by members of the public, rather than city officials. In the right plot of Figure 2, we calculate the tone (positive or negative, dropping neutral) of our discussions separately by speaker as described in *Methods*, and show that speaking members of the public are much less supportive of data centers than officials. This pattern is clear—as more members of the public speak in a given meeting, overall discussions about data centers become much more negative. In contrast, meeting discussions led only by city officials without community input discuss data centers in a more positive way. As above, we find evidence that this tone differential is asymmetric—while discussions led by city officials are slightly more positive overall (nearly 25pp net positive tone), our meetings with the highest proportion of public speakers are overwhelmingly negative (approximately 60pp net negative tone).

We offer a first look into how cities are considering a critical issue in America's adaptation to generative artificial intelligence. Power requirements mean data centers are often and will continue to be concentrated around urban areas. City governments regularly oversee land use, zoning, and siting provisions relevant to data centers. Given these two features of data centers, the city is central to the long-term infrastructural future of the digital economy. Although our analysis is preliminary—as the growth of the data center sector is rapid and ongoing—we see two contributions from our results. First, we validate emerging evidence—especially work produced by journalists—about public opposition to data centers in communities across the United States. In general, we find that local residents who participate in local policymaking debates about data centers are cautiously optimistic about the potential for future economic development, but are overall more concerned about how data centers will negatively impact their communities.

Second, we find evidence of a sharp divide between city government officials and members of the public regarding data centers. While data centers may offer cities the opportunity to host critical infrastructure for an evolving economy, residents are the ones who will pay the most direct costs of these projects, reflecting a familiar urban governance tension between growth and stasis. Our results offer the most comprehensive evidence yet about how cities across the country are discussing these trade-offs in their own neighborhoods, and highlight the need for robust public deliberation in data center siting.

2 Methods

We begin with the local government meetings collected and transcribed by the *LocalView* project [17]. For each meeting, we identify the city the meeting was held in, geocode it, and supplement the meeting data with demographic data from the 2024 American Community Survey and urbanicity data from the 2020 Census. In Supplementary Figure 2, we map all cities in the original data and those cities where we identified meetings with a data center discussion. Supplementary Table 2 compares the demographics of cities that discuss data centers, cities that do *not* discuss data centers, the full set of cities in the *LocalView* data

(regardless of whether or not data centers are discussed), and all Census places in the United States. Demographically, the places that discuss data centers are quite similar to places that do not discuss data centers in the *LocalView* data. However, both types of places tend to be more urban, less white, and more highly educated compared to the average Census place in the U.S. These differences are slightly larger for places that discuss data centers. We further discuss the urbanicity of our cities in the *Supplemental Information*.

Although *LocalView* already provides meeting transcripts, which are generally of high quality, we improve upon these transcripts using a more complex audio-to-speech model. We begin by using a keyword search to identify meetings where data centers are discussed. This keyword search also forms the basis of the data for Figure 1 and Supplementary Table 1. For the 1,707 meetings mentioning data centers, we download the original audio file of the meeting and produce a high quality transcript using OpenAI's Whisper model [18, 19]. Because local meetings often cover many different topics, and for computational efficiency, we subset to only the part(s) of the meeting where data centers are discussed (≈10,000). We call these individual units "discussions." This process is described in detail in the *Supplemental Information*. For each discussion, we identify the topics mentioned from a set of fifteen manually defined topics, whether the speaker in that discussion was a member of the public or a city official, and whether the speaker was discussing the topic in a positive, negative, or neutral way. We classify a random sample of 1% of discussions manually, then build an LLM prompt with the manual classifications as examples. We pass the prompt, and all of the discussions, to Claude Haiku 4.5 (2025-10-01) for classification of the remaining transcripts. We validate the classifications of topic, tone, and speaker types in the *Supplemental Information*. Finally, we aggregate our data in several plots to the meeting level, so each variable is reported as a proportion of the data center part of the meeting.

3 Data Availability

The data that support the findings of this study are available from the Harvard Dataverse [20].

4 Code Availability

The code to replicate the findings of this study are available from the Harvard Dataverse [20].

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Author Contributions

A.R., M.R., and T.S. designed research; performed research; analyzed data; and wrote the paper.

Competing Interests

The authors declare that they have no competing interests.

Figure Captions

Caption for Figure 1

Mentions of data centers in city meetings are growing, and emphasize land use, infrastructure, utilities, and the environment. The left plot shows that attention to data centers in city meetings is increasing over time. The x-axis shows meetings grouped into three month bins since the beginning of 2023 (we filter our meeting sample to start in 2023 for visual clarity), and the y-axis shows the proportion of meetings in that period that mention each issue. We compare discussions of data centers to several other types of local infrastructure projects. The data in tabular form, going back to 2009, can be found in Supplementary Table 1. In the right plot we show the share of meetings about data centers dedicated to each topic. The x-axis shows the share of meetings in which the topic was mentioned. We group the topics into five themes for clarity.

Caption for Figure 2

Cities discuss most topics negatively and members of the public are in general more negative than city officials. We use our tone classifications described in *Methods* to calculate, for each meeting-topic, the proportion of positive and negative discussions. The left plot shows the average difference between positive and negative proportions across all meetings (bars show 95% Wald CIs). In the right plot we show the average difference across all topics for members of the public versus city officials (bars show 95% Wald CIs). We categorize meetings in which data center discussions were made by only officials (0% public share, n=1,211), mostly officials (less than 25% public share, n=98), mixed (between 25% and 75%, n=147), and mostly public (greater than 75%, n=217).

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1 Supplementary Figures and Tables

1.1 Strings Used in Figure 1

We search transcripts to identify meetings about data centers, drug treatment clinics, casinos, and sports arenas. We search for the following strings to identify meetings falling into each of the categories: **data center:** `data center, data centre, server farm`, **Drug treatment centers:** `drug treatment clinic, drug treatment center, opioid treatment clinic, opioid treatment center`, **Casinos:** `casino`, **Sports arenas:** `sports stadium, sports arena`.

1.2 Attention to Data Centers over Time

Table 1: Quarterly share of all meetings mentioning each topic.

Quarter	Total meetings	Data centers	Casinos	Drug treatment clinics	Sports arenas
2009 Q1	123	0.0%	4.9%	0.0%	0.0%
2009 Q2	355	0.0%	0.6%	0.0%	0.0%
2009 Q3	202	0.0%	0.0%	0.0%	0.0%
2009 Q4	388	0.0%	0.3%	0.0%	0.5%
2010 Q1	470	0.2%	0.2%	0.0%	0.0%
2010 Q2	383	0.5%	0.3%	0.0%	0.0%
2010 Q3	435	0.0%	0.7%	0.0%	0.0%
2010 Q4	416	0.0%	0.7%	0.0%	0.0%
2011 Q1	944	0.0%	1.2%	0.0%	0.0%
2011 Q2	665	0.0%	2.0%	0.0%	0.0%
2011 Q3	653	0.2%	1.7%	0.0%	0.0%
2011 Q4	830	0.4%	1.1%	0.0%	0.0%
2012 Q1	1,117	0.3%	2.0%	0.0%	0.0%
2012 Q2	1,174	0.3%	1.5%	0.0%	0.1%
2012 Q3	1,067	0.7%	1.5%	0.1%	0.1%
2012 Q4	1,199	0.4%	1.3%	0.0%	0.1%
2013 Q1	1,719	0.5%	1.6%	0.0%	0.1%
2013 Q2	1,816	0.4%	1.5%	0.0%	0.1%
2013 Q3	2,019	0.3%	1.5%	0.0%	0.0%
2013 Q4	2,727	0.1%	1.4%	0.0%	0.1%
2014 Q1	2,659	0.1%	1.4%	0.0%	0.0%
2014 Q2	3,090	0.2%	1.2%	0.0%	0.1%
2014 Q3	2,858	0.2%	1.2%	0.0%	0.1%
2014 Q4	3,104	0.3%	1.0%	0.0%	0.1%
2015 Q1	3,385	0.4%	1.2%	0.0%	0.1%
2015 Q2	3,504	0.3%	1.5%	0.0%	0.1%
2015 Q3	3,570	0.2%	1.0%	0.0%	0.1%
2015 Q4	3,721	0.2%	0.9%	0.0%	0.1%
2016 Q1	4,196	0.1%	1.7%	0.0%	0.1%
2016 Q2	4,343	0.2%	0.8%	0.0%	0.1%
2016 Q3	4,437	0.4%	0.9%	0.0%	0.1%
2016 Q4	4,009	0.1%	1.0%	0.0%	0.1%
2017 Q1	4,603	0.2%	0.7%	0.0%	0.0%
2017 Q2	5,234	0.5%	2.1%	0.0%	0.2%
2017 Q3	5,459	0.7%	3.8%	0.0%	0.2%
2017 Q4	4,745	0.5%	1.7%	0.0%	0.3%
2018 Q1	5,565	0.6%	2.2%	0.0%	0.2%
2018 Q2	6,381	0.8%	2.0%	0.0%	0.1%

Table 1: Quarterly share of all meetings mentioning each topic. *(continued)*

Quarter	Total meetings	Data centers	Casinos	Drug treatment clinics	Sports arenas
2018 Q3	5,795	0.5%	1.7%	0.0%	0.2%
2018 Q4	5,473	0.8%	1.6%	0.0%	0.1%
2019 Q1	6,038	0.7%	2.3%	0.0%	0.2%
2019 Q2	6,459	0.5%	1.9%	0.0%	0.2%
2019 Q3	5,618	0.4%	2.0%	0.0%	0.1%
2019 Q4	6,313	0.6%	2.0%	0.0%	0.1%
2020 Q1	7,574	0.3%	1.9%	0.0%	0.1%
2020 Q2	6,297	0.4%	2.4%	0.0%	0.1%
2020 Q3	6,462	0.5%	1.7%	0.0%	0.2%
2020 Q4	6,379	0.6%	1.8%	0.0%	0.1%
2021 Q1	7,384	0.7%	1.7%	0.0%	0.1%
2021 Q2	7,724	0.7%	1.6%	0.0%	0.1%
2021 Q3	6,632	0.5%	1.8%	0.0%	0.2%
2021 Q4	6,293	0.7%	1.7%	0.0%	0.1%
2022 Q1	7,203	0.6%	2.4%	0.0%	0.2%
2022 Q2	7,078	0.8%	2.0%	0.1%	0.2%
2022 Q3	6,364	0.7%	2.4%	0.0%	0.2%
2022 Q4	6,258	0.7%	2.0%	0.1%	0.2%
2023 Q1	7,052	0.7%	2.7%	0.0%	0.2%
2023 Q2	7,097	0.8%	2.3%	0.0%	0.2%
2023 Q3	6,276	0.9%	2.7%	0.0%	0.1%
2023 Q4	6,415	0.7%	1.9%	0.0%	0.2%
2024 Q1	6,994	0.8%	2.1%	0.0%	0.2%
2024 Q2	7,452	1.2%	2.1%	0.0%	0.2%
2024 Q3	6,579	1.3%	2.1%	0.0%	0.2%
2024 Q4	6,737	1.9%	2.0%	0.0%	0.2%
2025 Q1	7,409	2.4%	1.9%	0.0%	0.2%
2025 Q2	7,990	2.2%	2.0%	0.1%	0.2%
2025 Q3	6,990	3.7%	2.1%	0.0%	0.2%
2025 Q4	7,171	4.4%	1.9%	0.0%	0.2%
2026 Q1	7,385	6.8%	1.9%	0.0%	0.2%

1.3 Places Discussing Data Centers

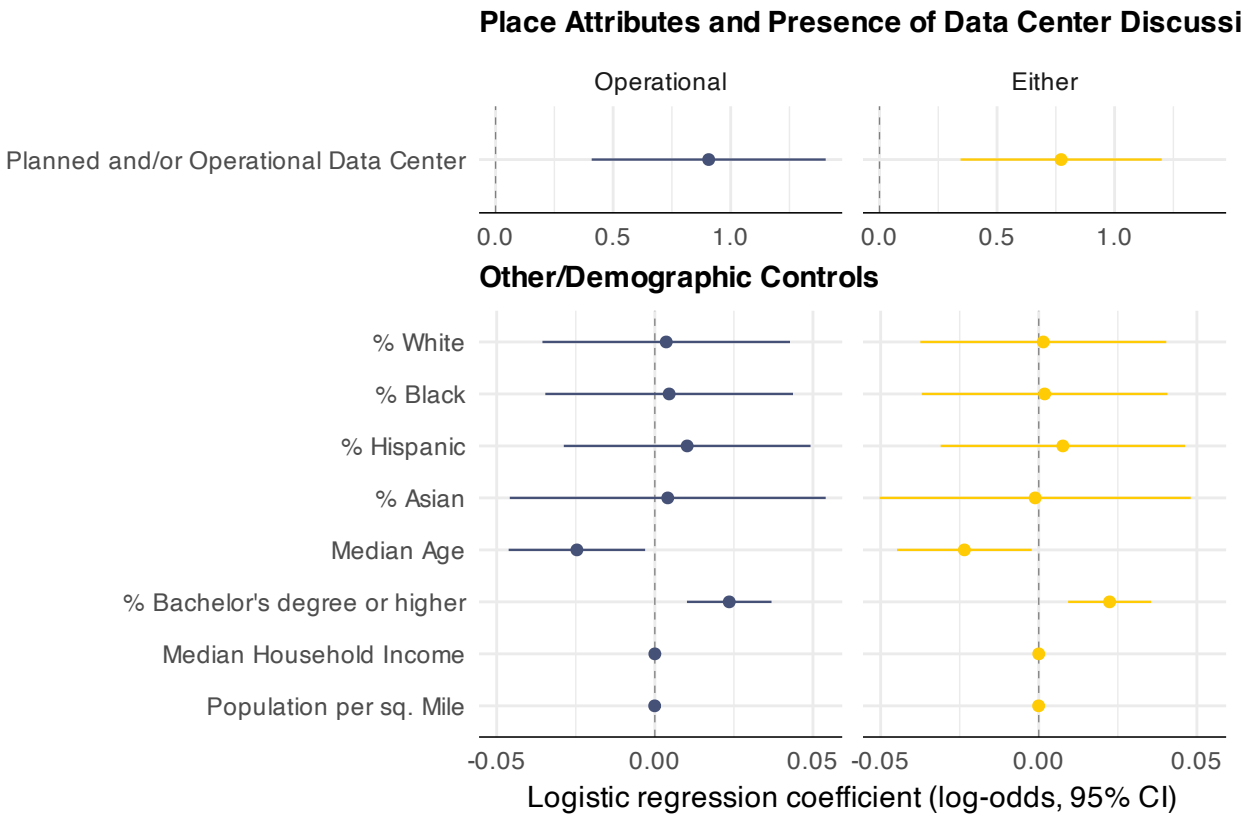


Figure 1: **Data Center Discussion Model Coefficients.** Logistic regression coefficients (log-odds, with 95% confidence intervals) for models predicting whether a place discussed data centers at local government meetings. The top panel shows the coefficient on whether the place hosts or plans to host a data center, under two configurations. The first model compares places with an operational data center (and no planned data centers) to places with no planned or operational data centers. The second model compares places with either a planned or operational data center to places with neither. The bottom panel shows coefficients for the same models' covariates. Model sample sizes are 869 and 905, respectively. The figure provides empirical support for the intuitive idea that places with planned or operational data centers are more likely to discuss them in meetings, compared to localities that do not. While data centers are more likely to be discussed when they are a concrete and local concern, we still find many places that discuss data centers without a current or planned project in their community. The number of places that have planned and/or operational data centers in our sample is fairly small: 135 places have either a planned or operational data center, and 99 places have operational data centers with no additional planned data centers. 770 places had no planned or operational data centers. Unfortunately, we are unable to identify timing of data center opening or construction. For instance, we cannot track the development of a specific data center over time, or identify if discussions at a particular meeting coincide with the proposal of a specific data center.

1.4 Urban Demographics of the Data

Of the 905 sample localities matched to a Census geography, 91 (10.1%) are big cities, 249 (27.5%) are smaller cities, 413 (45.6%) are suburbs, and 152 (16.8%) are rural. Localities are classified as urban when at least half their 2020 population lived in an urban area, and are then split at the 60th and 90th population percentiles of the sample (27,278 and 144,187 residents); urban localities below the 60th percentile are labeled suburbs. Examples include El Paso, TX and Albuquerque, NM (big cities), New Milford, CT and Plainfield, NJ (smaller cities), Caruthersville, MO and Cochran, GA (suburbs), and Oakland, ME and Macon County, NC (rural).

Supplementary Figure 2 maps each locality in our sample in maize. The figure also includes, in black, localities that are in the *LocalView* data but do not have a recorded discussion of data centers in our data.

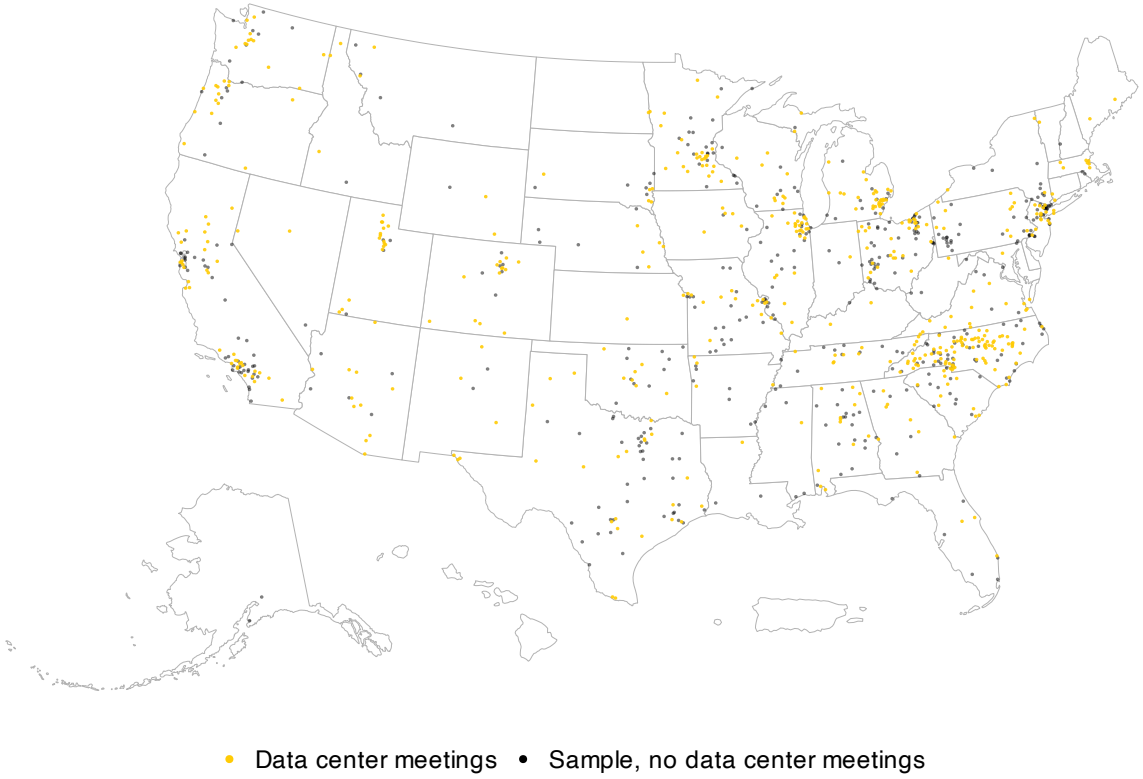


Figure 2: **Map of Meeting Locations**. A map of all localities in the *LocalView* data (black dots) and all localities with a mention of a data center in the entire study period (maize dots).

1.5 Main Results by Urbanicity

In this subsection, we reproduce each of Figure 1a, Figure 1b, Figure 2a, and Figure 2b of the main text, but break the figures down by the urbanicity of the locality. We use the same distinction defined above, i.e., into big cities, smaller cities, suburbs, and rural places. As can be seen in the figures, our main results are relatively stable across city types.

Table 2: **Demographic comparison of localities with data center meetings, all sample localities, and all U.S. incorporated places (2024 ACS 5-year estimates)**. The top panel weights each locality equally and reports the median across units (the typical *place*); the bottom panel is population-weighted, so shares are pooled totals and age, income, and density are population-weighted means (the typical *resident*).

Measure	Localities with data center meetings	All sample localities	All U.S. places
Localities (N)	469	905	32,038
The typical place (equal-weighted median)			
White (non-Hispanic)	69.9%	70.5%	83.3%
Black (non-Hispanic)	4.5%	4.1%	0.4%
Hispanic	9.0%	8.2%	4.1%
Asian (non-Hispanic)	1.9%	1.6%	0.0%
Median age	38.6	39.0	41.1
Bachelor's degree or higher	33.8%	29.9%	20.6%
Median household income	$73,750	$72,083	$67,514
Population per sq. mile	1,687	1,570	700
The typical resident (population-weighted)			
White (non-Hispanic)	50.7%	47.6%	52.0%
Black (non-Hispanic)	14.5%	14.9%	13.2%
Hispanic	24.2%	26.8%	22.2%
Asian (non-Hispanic)	5.3%	5.6%	6.9%
Median age	36.3	36.4	38.1
Bachelor's degree or higher	39.0%	38.2%	36.8%
Median household income	$80,248	$80,272	$85,898
Population per sq. mile	3,612	3,499	4,437

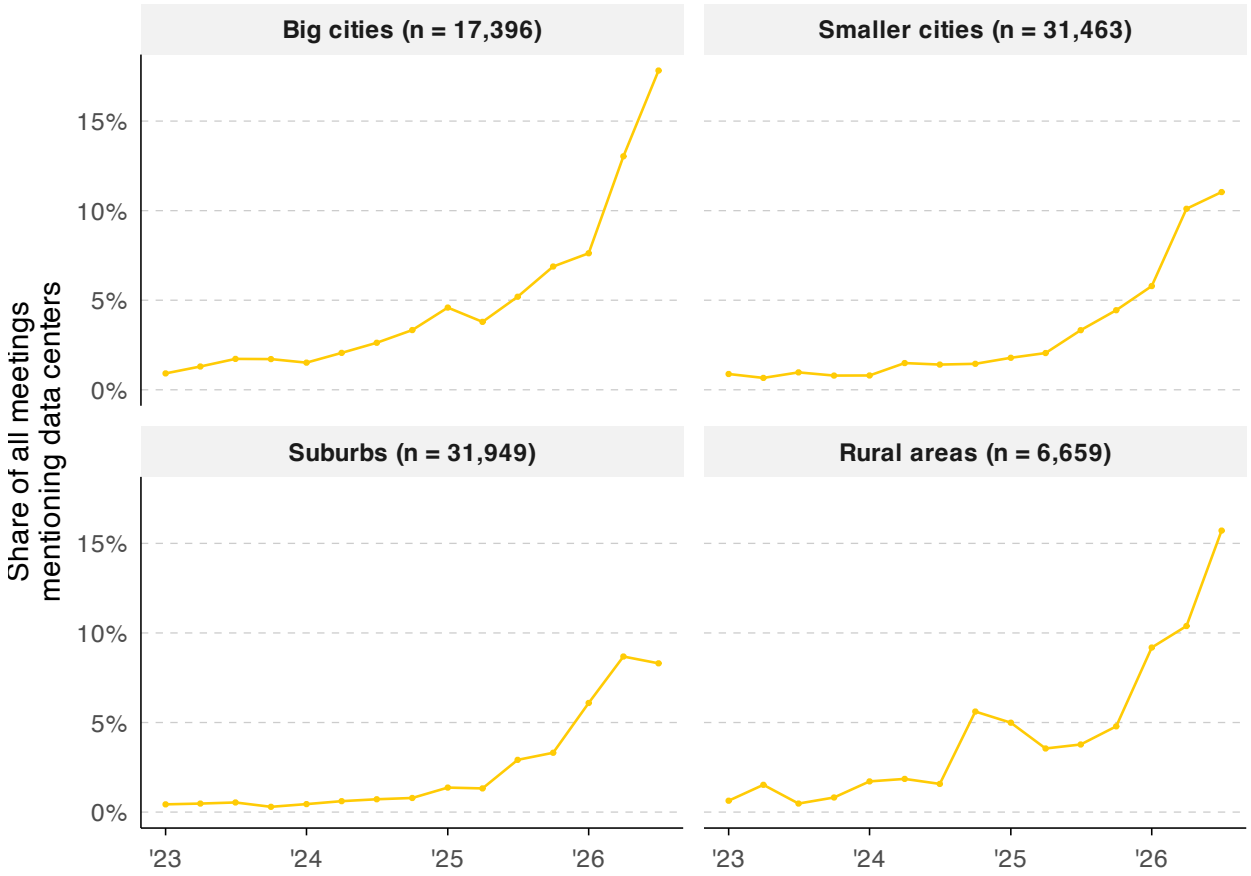


Figure 3: Figure 1a, by urbanicity

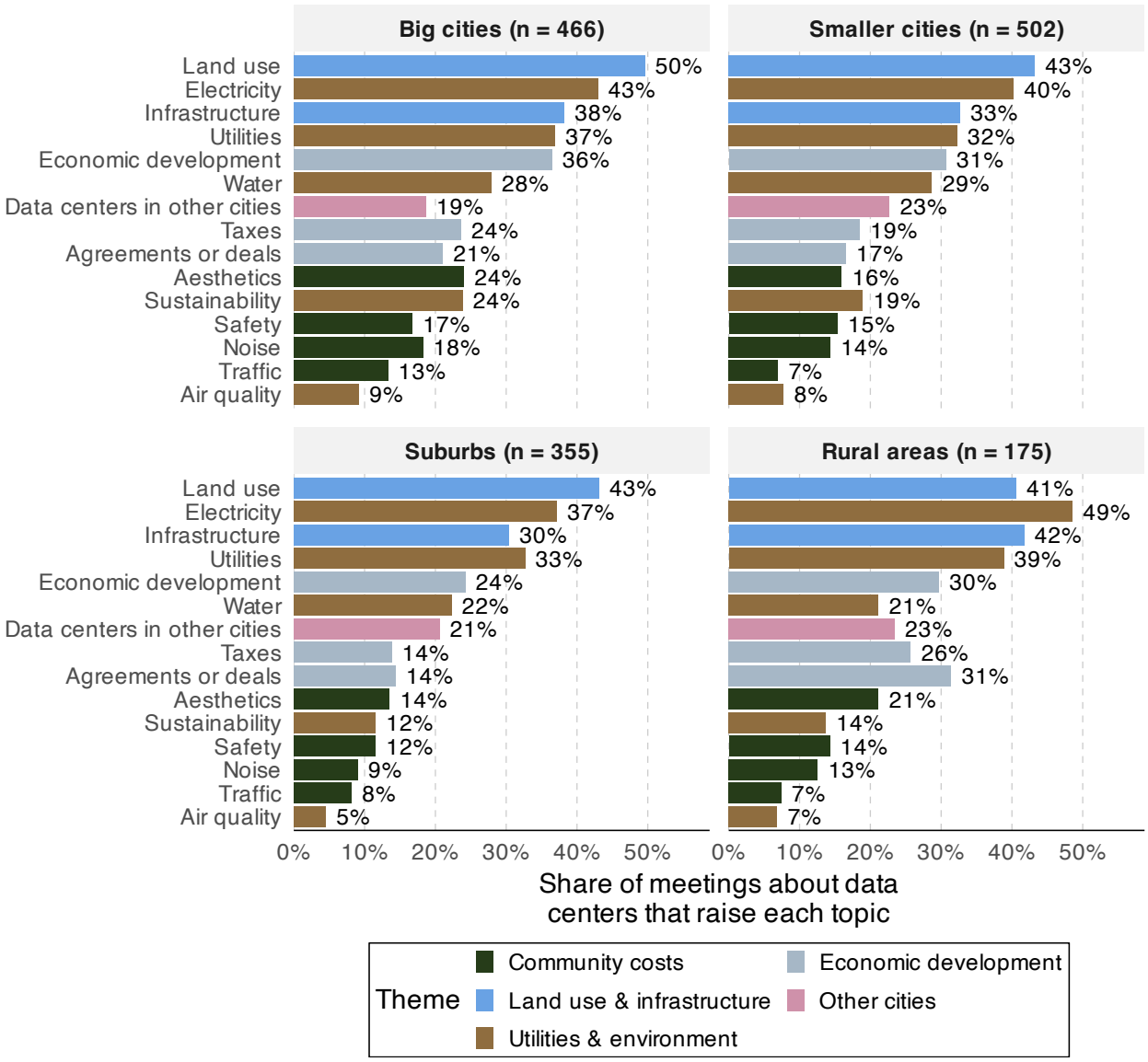


Figure 4: Figure 1b, by urbanicity

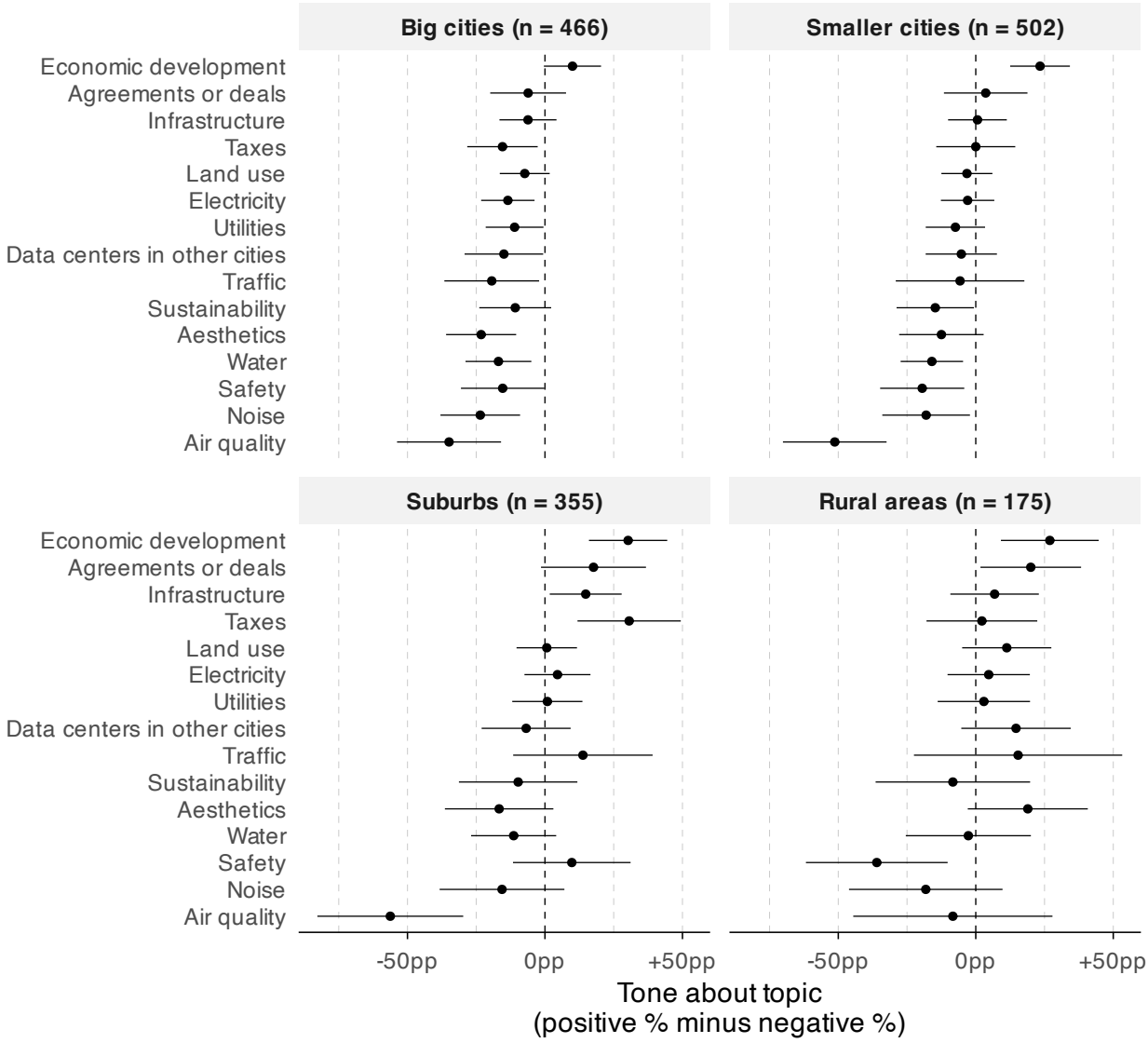


Figure 5: Figure 2a, by urbanicity

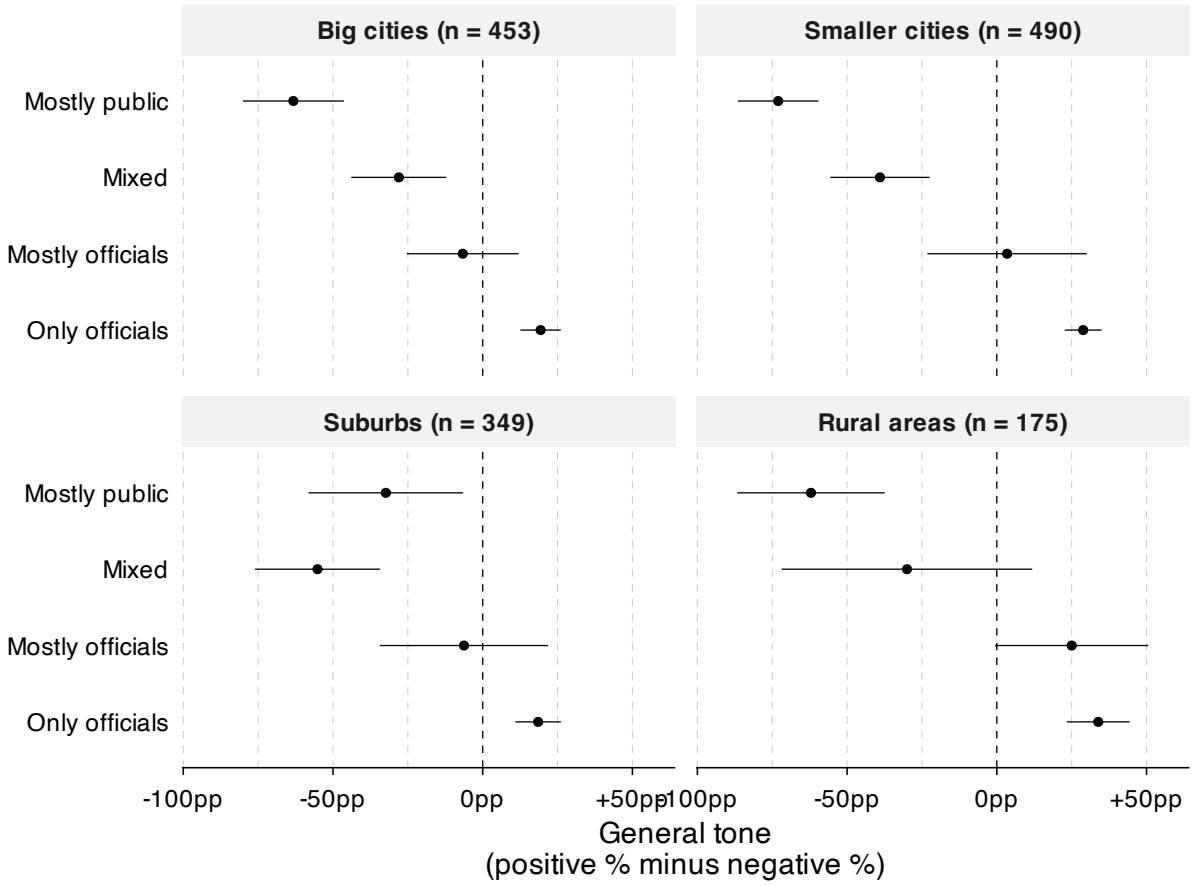


Figure 6: Figure 2b, by urbanicity

2 LLM Prompts and Validation

As discussed in the main text, we identify all meetings in our dataset that feature a discussion of data centers using keyword searches. Meeting transcripts were preprocessed by collapsing extra whitespace, then segmented into sentences using a punctuation-based split that accounts for common abbreviations (e.g., Dr., vs., U.S.) to avoid false sentence boundaries. Sentences were then grouped into fixed-size blocks of ten consecutive sentences, which served as the unit of analysis. In the main paper, we refer to these chunked sections of meetings as "discussions." Then, we manually coded 128 of these chunks that discuss data centers for theme, tone, and speaker information. These were then provided, along with detailed prompts, as training information for an LLM to code the roughly 10,000 total chunks. The LLM coded each chunk for thematic content, tone, and identified information about the speaker. In this section, we provide some validation statistics, showing the level of agreement between our manually coded observations and the results of the same discussions passed to the LLM. We also include the prompts provided to the LLM for each. We used Claude Haiku 4.5 (2025-10-01) for all LLM tasks.

2.1 LLM Theme Coding Prompt

The following prompt was given to Claude for coding themes:

For each topic below, code 1 if it was discussed **at any point** in the section of the meeting, and 0 if it was not mentioned at all.

- *Environment*
  - `any_water`: Water use, wastewater, cooling, sewer, runoff, or water cost.
  - `any_electricity`: Energy demand of the data center, power grid references, or electricity cost.
  - `any_air_quality`: Air quality concerns or pollution.
  - `any_sustainability`: Sustainability or climate concerns.
  - `any_utility`: Generic public utility references (other than water or electricity), or utility cost.
- *Community Costs*
  - `any_noise`: Noise from the data center project (noise pollution, construction, server hum).
  - `any_traffic`: Traffic during construction or operation.
  - `any_safety`: Safety concerns: hazmat, waste runoff, fire risk, electromagnetic concerns, local emergency response capacity.
  - `any_infrastructure`: General infrastructure concerns related to data centers.
  - `any_aesthetics`: Neighborhood character, visual or aesthetic impacts.
  - `any_equity_concerns`: Equity in the siting process, or distribution of benefits and costs across residents.
- *Economic Impacts*
  - `any_economic_development`: Jobs, local hiring, pay, unionization, or number of positions.
  - `any_taxes`: Tax or fiscal benefits/burdens of the data center on the community.
  - `any_land_use`: Productivity or appropriateness of land use for a data center vs. alternatives.
  - `any_deal`: The agreement with the developer: cost sharing, tax incentives, infrastructure responsibilities.

2.2 LLM Speaker Coding Prompt

The following prompt was given to Claude for coding speaker information:

For each section, identify the primary category of speaker. Code 1 if the section is primarily spoken by that speaker, and 0 otherwise. Use your best judgment based on the transcript. Speaker categories are **mutually exclusive**:

- **Developer** (`developer`): Speech by a representative of the company proposing or building the data center—a developer, consultant, or lawyer speaking on the applicant's behalf.
  - *Example*: "I am here tonight to speak on behalf of Crytalis Developers..." or "As the project engineer for the applicant..."
- **Public** (`public`): Speech by a member of the public—not an elected/appointed official or a developer representative.
  - *Example*: "My name is Tyler Simko, and I strongly oppose this data center project on 5th..."
- **Official** (`official`): Speech by a formal local official—elected or appointed government staff.
  - *Example*: "I have always said I am a no on this project. I am going to vote no because of the extensive feedback I have heard..."
- **Other** (`other`): Speech by someone who fits none of the above—for example, an invited expert witness, a state agency representative, or an unidentifiable speaker.
  - *Example*: "As the state environmental liaison, I'm here to summarize the permitting requirements..."

2.3 LLM Tone Coding Prompt

The following prompt was given to Claude for coding tone:

Code which tones are present at any point in the section of the meeting. Multiple tone flags can be 1 simultaneously—a meeting may contain both supportive and critical speech. Each flag is independent.

- `any_positive_tone` (1 vs. 0): Was any portion of the section clearly supportive of the data center project, or speaking positively about data centers in general? The dominant thrust of that portion must be favorable.
- `any_negative_tone` (1 vs. 0): Was any portion of the section clearly opposed to the data center project, or speaking critically about data centers in general? The dominant thrust of that portion must be critical or opposed.
- `any_neutral_tone` (1 vs. 0): Was any portion of the section neutral—factual, procedural, or not taking a clear stance? Purely administrative segments (vote tallies, scheduling, introductions) always count as neutral.

Default rule: When a section contains mixed signals with no clear dominant thrust, count it as neutral, not positive or negative.

2.4 Validation of LLM Theme Coding

Manual and LLM codings are considered to be in agreement if values for the same section match. Using these matches, we then aggregate over all 128 sections and calculate the average number of same classifications.

Agreement between manual and LLM-assigned codes was generally high across thematic variables, though with some variation. Among environmental indicators, agreement was 95.31% for water, 85.16% for electricity, 96.09% for air quality, 86.72% for sustainability, and 84.38% for utility. Community indicators showed similarly strong agreement, with noise at 96.88%, traffic at 96.88%, safety at 91.41%, infrastructure at

75.00%, aesthetics at 85.94%, and equity concerns at 84.38%. The agreement value for infrastructure—noticeably lower than the rest—is likely driven by our rather generic categorical description referencing "general" infrastructure concerns. Our intention was to treat the infrastructure category as a way to capture other miscellaneous concerns regarding the potential costs on communities of hosting this infrastructure that are not captured by other categories. However, this lack of clarity likely drives the low result. Agreement on economic impact variables was 89.06% for economic development, 95.31% for taxes, 81.25% for land use, and 83.59% for deal.

We also compare the overall distribution of topics between the manual coding and the LLM codings. Supplementary Figure 7 compares the share of discussions in which the topic was mentioned. Generally speaking, the LLM and manual coded discussions have quite similar topical distributions.

2.5 Validation of LLM Speaker Categorization

We performed a similar validation exercise for speaker categorization. Speakers were either categorized as an elected or formal local official, a member of the public, a representative from a data center developer, or someone else.

Agreement on speaker categorization was more mixed across the four categories. Agreement was 78.12% for "developer", 81.25% for "public", 74.22% for "official", and 87.50% for "other". In our main text results, we use only the public and official categories.

2.6 Validation of LLM Tone Categorization

Sections were coded as either Positive, Negative, or Neutral. Agreement between manual and LLM classification on tone was 75.78% overall. However, this overstates the level of disagreement between the manual and LLM codings because many of these errors were driven by discrepancies in the "neutral" category, which we do not use in our analyses. Directional agreement between positive and negative codings was much stronger. Of the 50 observations coded as negative manually, only 3 were classified as positive by the LLM. Of the 17 observations manually coded as positive, only 1 was coded as negative by the LLM.

We also report the tone by each topic using only the manual classifications in Supplementary Figure 8. Despite the smaller sample sample, the results in this figure match those shown in Figure 2 of the main text.

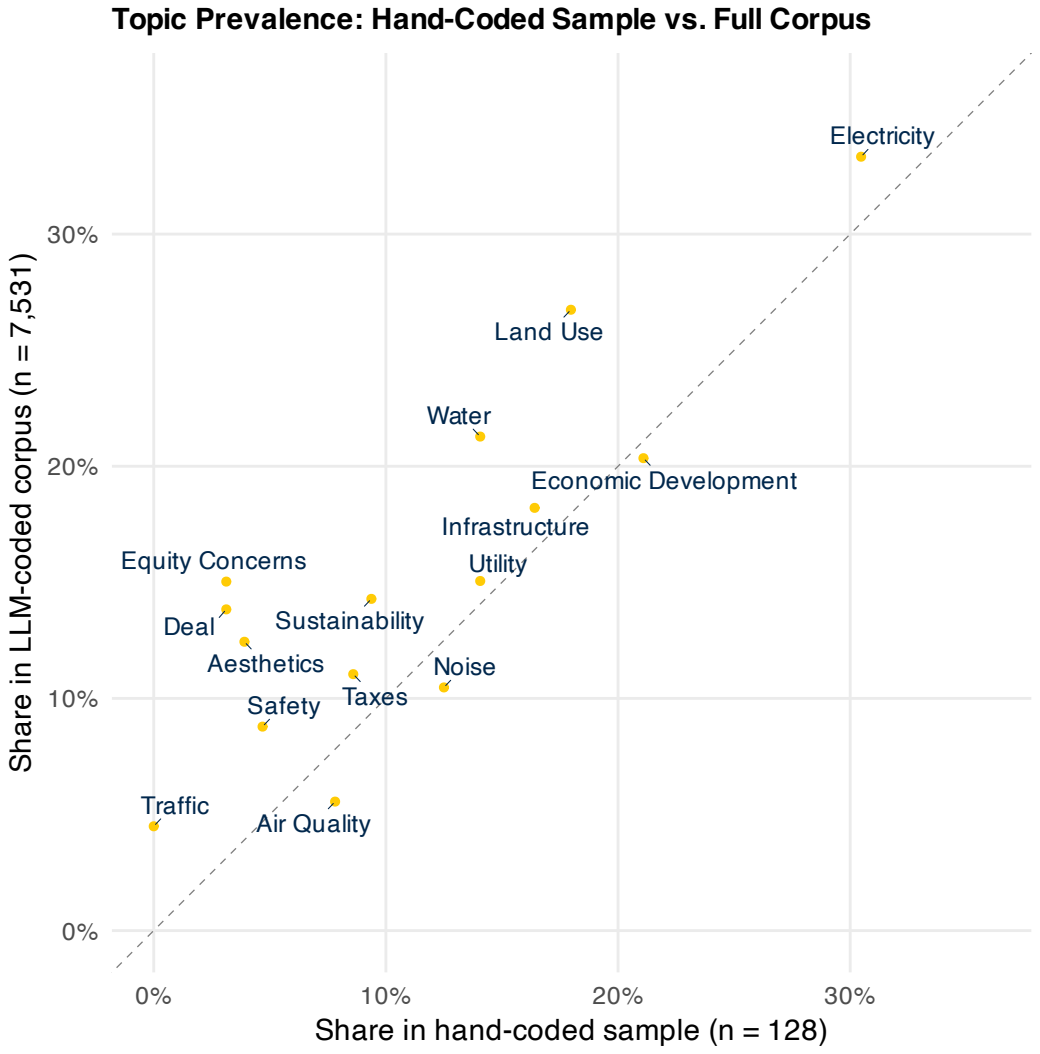


Figure 7: **LLM-assigned versus Manually-assigned topic codes.** For each of the substantive topics, we compare the share of manually coded discussions in which the topic was mentioned (x-axis, $n = 128$) to the share of the full LLM-coded corpus in which the topic was mentioned (y-axis, $n = 7{,}531$). Points falling near the dashed 45-degree line indicate agreement between the manual and LLM-coded prevalence of each topic. Generally speaking, we see that the LLM and manual coded discussions have quite similar topical distributions.

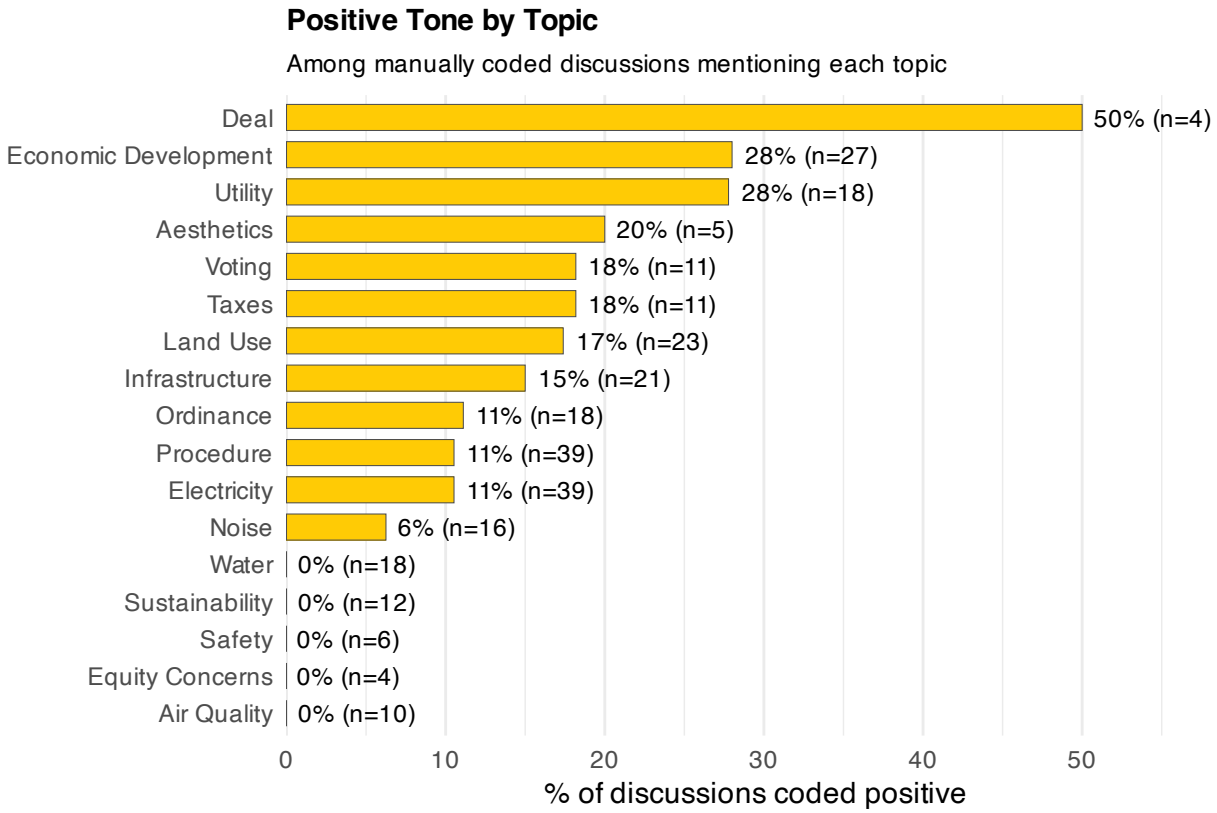


Figure 8: **Manually Coded Discussions by Topic and Tone.** The figure plots the share of manually coded discussions that had a positive tone by topic. The figure also shows the number of discussions mapped to each topic. Despite the small sample size, the results match well with those shown in Figure 2a in the main text.